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iCE40UP5K-B-EVN



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Mouser No:
842-ICE40UP5K-B-EVN

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Mfr.:
[Lattice](#)

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Description:
Programmable Logic IC Development Tools ICE40 ULTRAPLUS BRD 48QFN I/O ACCESS

Datasheet:
[iCE40UP5K-B-EVN Datasheet](#)

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Factory Lead Time:
2 Weeks 

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Specifications

All Products > Embedded Solutions > Engineering Tools > Sensor Development Tools > Multiple Function Sensor Development Tools >

SparkFun SEN-12650

SEN-12650



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Mouser No:
474-SEN-12650

Mfr. No:
SEN-12650

Mfr.:
[SparkFun](#)

Customer No:

Description:
Multiple Function Sensor Development Tools The AD8232 SparkFun Single Lead Heart Rate Monitor is a cost-effective board used to measure the electrical activity of the heart. This electrical activity can be charted as an ECG or Electrocardiogram and output as an analog reading. ECGs can be ext

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Specifications

Product Attribute	Attribute Value	Select Attribute
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EMBEDDED EDGE INDUSTRIAL SYSTEM

HUMAN BODY
(3-Lead Electrode Pads)

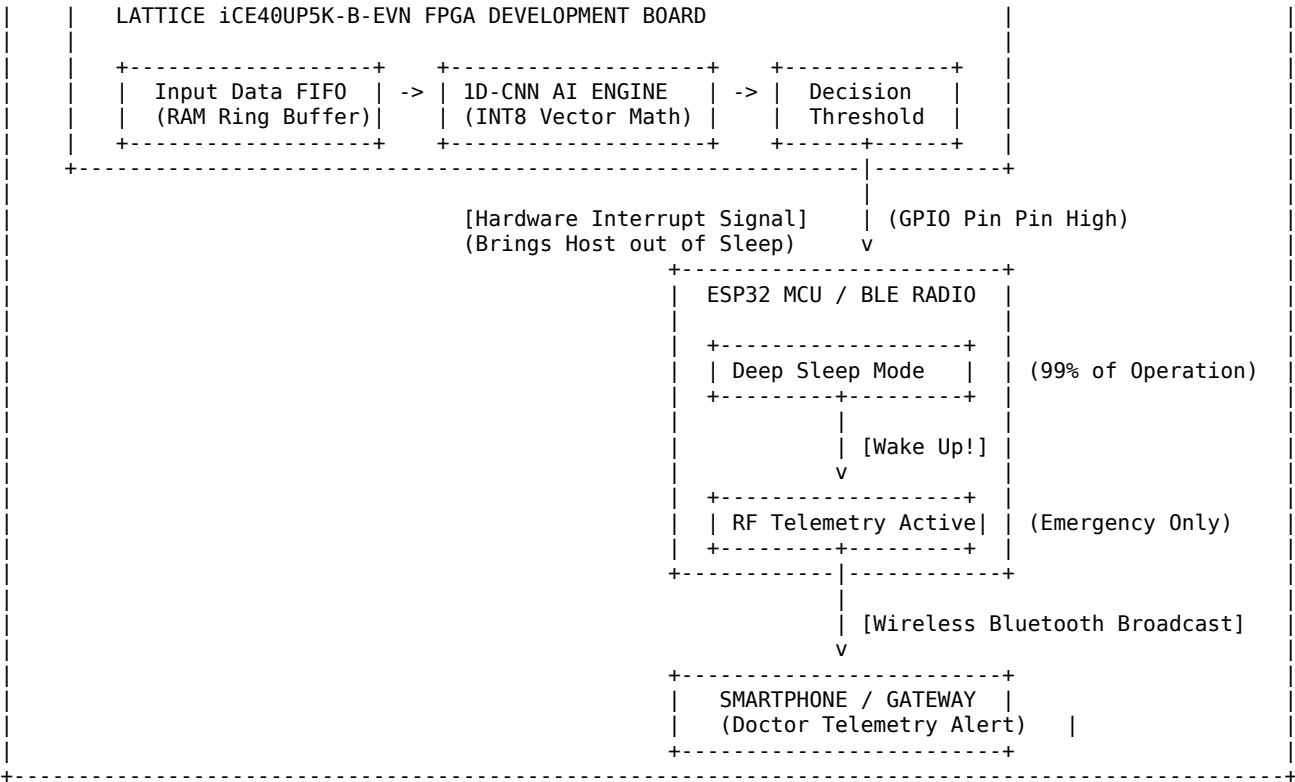
[Analog Bio-Potentials]

AD8232 ECG SENSOR
(Analog Front End)

[Filtered Analog Wave]

ADS1115 ADC MODULE
(Analog-to-Digital)

[SPI / I2C Bus Data Streams]



The Final Shopping List

To implement this architecture completely, you must add these precise items to your shopping cart alongside your FPGA board.

1. Sensors & Translators

- AD8232 Single-Lead Heart Rate Monitor Kit:** Ensure the kit explicitly includes the evaluation module board, the 3-pin sensor cable, and at least 3 to 10 disposable **Ag/Cl biomedical electrode pads**.
- ADS1115 16-Bit I2C ADC Module:** A high-precision analog-to-digital breakout board with an integrated voltage reference. This translates the smooth voltage waves coming out of the AD8232 sensor into numeric digital data for the FPGA.

2. Alert & Telemetry Engine

- ESP32-WROOM-32D Development Board:** (NodeMCU style). This micro-controller provides an ultra-low-power deep sleep mode along with integrated Bluetooth Low Energy (BLE). It serves as your wireless transmitter, remaining asleep until triggered by the FPGA's interrupt pin.

3. Lab Infrastructure & Interconnects

- 1x Standard 830-Point Solderless Breadboard:** Serves as the temporary physical platform to share power rails and secure data lines between your sensor, ADC, FPGA, and ESP32.
- 65-Piece Assorted Male-to-Male (M-M) Jumper W-Wires:** For making physical node connections across the channels of your breadboard.
- 40-Pin Female-to-Female (F-F) Jumper Wires:** Vital for making robust connections directly from the bare header male pins of the iCE40UP5K board straight down into your breadboard nodes.
- 1x Premium Mini-USB Cable:** Ensure it handles data transmission, not just power. The iCE40UP5K-B-EVN uses an older **Mini-USB Type-B** hardware receptacle to flash bitstreams from your computer.

Phase 1: Bench Prototyping (On Your Desk)

The goal here is to make the entire hardware loop talk to each other without errors.

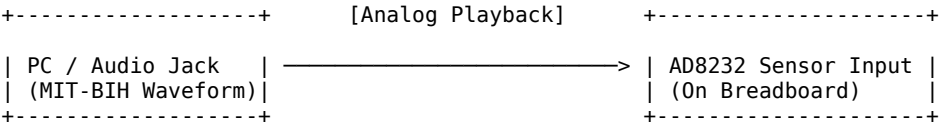
[Patient / Simulator] -> [AD8232 Sensor] -> [ADS1115 ADC] -> [iCE40 FPGA Core (CNN)] -> [Interrupt Pin] -> [ESP32 BLE] -> [Phone]

- The Software Framework:** Build and train the 1D-CNN in Python using the **MIT-BIH Arrhythmia Database**. Apply Quantization-Aware Training (QAT) to output an INT8 model.
- The Hardware Logic:** Write the structural C/C++ logic for the CNN layers. Use a High-Level Synthesis (HLS) workflow to output generic Verilog/VHDL code, then compile it inside **Lattice Radiant** to program the **iCE40UP5K-B-EVN** board.

3. **The Interconnect Circuit:** Mount the AD8232 sensor and the ADS1115 ADC onto your breadboard. Wire the ADC to the FPGA using an I2C/SPI digital bus. Connect a single GPIO pin from the FPGA to an input pin on the ESP32 to act as the wake-up wire.
4. **The Radio Logic:** Program the ESP32 to enter its ESP_PD_DOMAIN_RTC_PERIPH Deep Sleep state. Configure it to wake up instantly only when that single FPGA wire transitions from LOW to HIGH.

Phase 2: Pre-Clinical Verification (Safety First)

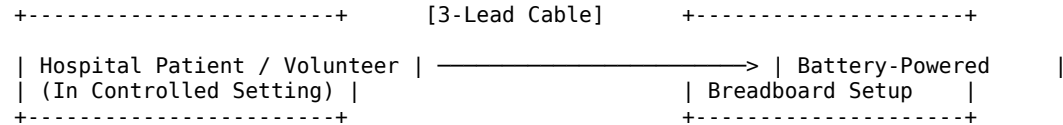
Crucial Step: You cannot safely attach a homemade, wall-powered breadboard prototype directly to a living human being. If your computer's power supply spikes, it could pass a dangerous electrical current through the electrode pads straight across the patient's chest.



1. **Isolate the System:** Disconnect your laptop from the wall and run the entire prototype strictly on a **5V USB Power Bank (Battery Power)**. This ensures complete isolation from mains electrical hazards.
2. **Use a Signal Simulator:** Before testing on humans, convert an ECG record from the MIT-BIH database into an audio wave format (.wav). Play it out of your computer's headphone jack using an auxiliary cable connected directly into the AD8232 sensor's input pins.
3. **Validate the Logic:** Verify that when a normal heartbeat plays, the system stays quiet. When an irregular heartbeat plays, ensure the FPGA pin flips high, waking up the ESP32 to transmit the alert to your smartphone.

Phase 3: Clinical / Field Testing (The Hospital Phase)

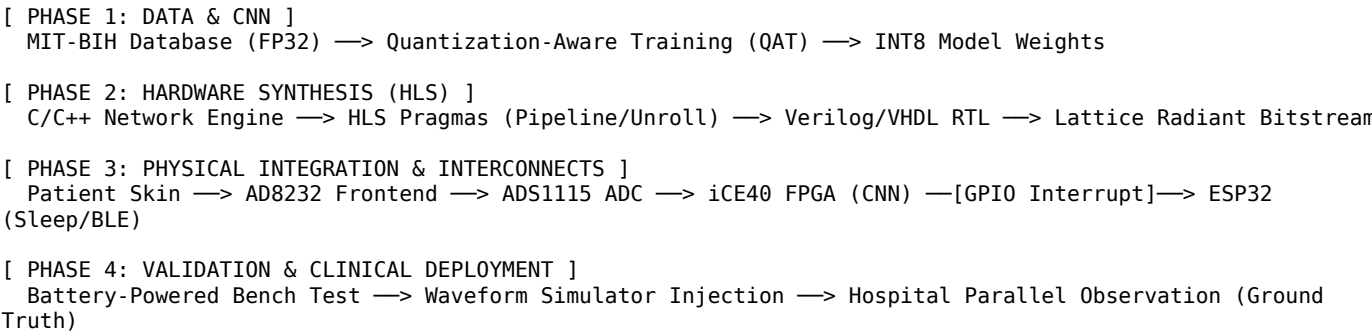
Once your simulator tests achieve high accuracy, you can move toward field testing.



1. **Secure Institutional Approval:** To test this in an actual medical facility, you must coordinate with a consulting cardiologist and obtain formal permission (or an Institutional Review Board approval) to run an observational trial.
2. **The Clinical Protocol:** Keep the hospital's certified medical-grade ECG monitor attached to the patient as the official ground truth. Concurrently, attach your isolated, battery-powered 3-lead AD8232 gel pads to the patient's limbs or chest.
3. **Observation & Logging:** Run your setup side-by-side with the hospital monitor. When the hospital monitor flags an event (like a premature ventricular contraction), verify if your FPGA successfully registered the anomaly, fired the hardware interrupt, and logged the event on your phone.

This document serves as your **Master Technical Reference Blueprint** for the event-driven Edge AI ECG Monitor. Keep this as a reference guide to maintain the correct software-to-hardware intuition at every step of your development cycle.

Architecture Reference Map



Phase 1: Data Architecture & Network Prototyping (Python)

Objective

To build a highly accurate, heavily compressed Neural Network in Python that learns the unique shapes of normal vs. abnormal heartbeats, keeping the total size under the **15 KB memory limit** of the iCE40UP5K FPGA.

Key Technical Operations

- **Signal Denoising:** Raw ECG signals contain severe muscle tremors and breathing drift. You must apply a digital high-pass filter ((0.5 Hz)) to anchor the baseline and a notch filter ($(50/60\text{ Hz})$) to remove electrical wall-hum.
 - **R-Peak Segmentation:** Use the Pan-Tompkins algorithm to locate the sharpest point (R-peak) of every heartbeat. Slice a fixed-size window of **90 data points** centered directly on that peak. This window isolates the complete P-QRS-T wave structure.
 - **Quantization-Aware Training (QAT):** Standard neural networks use large 32-bit floating-point numbers (FP32). The FPGA cannot calculate these efficiently. You must train the model using QAT via PyTorch or TensorFlow. This processes the math in 8-bit integers (INT8) during training, forcing the network to maintain high accuracy despite the severe data compression.
 - **The Weight Budget:** The architecture must contain fewer than 15,000 total weights to fit into the FPGA's local Block RAM. Keep the network restricted to two 1D-Convolutional layers (4 to 8 filters each) and one tiny Fully-Connected decision layer.
-

Phase 2: Hardware Architecture Translation (C/C++ to RTL)

Objective

To convert your trained Python network into fixed-point hardware circuitry using High-Level Synthesis (HLS), ensuring that calculations take place directly on silicon without an operating system.

Key Technical Operations

- **C/C++ Structural Modeling:** Rewrite the internal loop structures of the CNN (Multiplication-Accumulation matrix math) into pure, structural C/C++. Avoid all dynamic variables; array sizes must be completely hardcoded.
 - **HLS Pragma Optimization:** Inject specialized compiler directives (pragmas) into your loops to command how the FPGA routes its physical logic:
 - `#pragma HLS PIPELINE`: Allows data points to enter the processing engine sequentially on consecutive clock cycles, preventing data jams.
 - `#pragma HLS UNROLL`: Duplicates physical hardware multipliers so multiple array values are multiplied simultaneously instead of one by one.
 - `#pragma HLS ARRAY_PARTITION`: Splits single memory blocks into separate physical registers, allowing the network to access all its weights in a single clock cycle.
 - **Bit-Growth Management:** Multiplying two 8-bit numbers results in a 16-bit number. Adding them up yields a 32-bit number. You must include explicit truncation logic at the end of each layer to safely scale and clip 32-bit accumulators back down to an 8-bit integer (INT8) before feeding the data into the next layer.
 - **Bitstream Generation:** Compile the generic Verilog/VHDL code generated by your HLS workflow inside **Lattice Radiant** to create the final .bin bitstream file, then flash it directly onto the iCE40UP5K-B-EVN board via its mini-USB port.
-

Phase 3: Physical Integration & Interconnects (The Breadboard)

Objective

To physically construct the complete device on a solderless breadboard, wiring the sensor, the data translator, the FPGA engine, and the sleeping radio transmitter together.

Key Technical Operations

- **The Analog Frontend Connection:** Place the AD8232 module on the breadboard. Connect its input to the 3-lead biomedical electrode pads. The AD8232 captures the heart's tiny microvolt signals, amplifies them, and outputs a clean, smooth analog voltage wave.
 - **The Data Translator (ADC):** Connect the analog output pin of the AD8232 straight into the analog input channel of the ADS1115 ADC. Wire the digital output pins of the ADS1115 into the input pins of the iCE40 FPGA using an I2C or SPI digital communication bus. This constantly feeds digitized voltage values into the FPGA.
 - **The Interrupt Wire:** Identify a free General Purpose Input/Output (GPIO) pin on the iCE40 FPGA and connect a jumper wire from it directly to an input pin on the ESP32 microcontroller. This is your critical **Hardware Interrupt Line**.
 - **The Sleeping Radio Logic:** Program the ESP32 to boot up, immediately drop its internal radio components into a deep power-down sleep state, and wait. Program its Real-Time Clock (RTC) wake-up logic to monitor that single incoming interrupt wire from the FPGA.
-

Phase 4: Validation & Clinical Testing (The Hospital)

Objective

To safely test your breadboard system, validating its ability to successfully capture anomalies and save battery power when exposed to simulated signals and real clinical environments.

Key Technical Operations

- **Total Battery Isolation:** **Never connect a wall-powered computer to a device attached to a human being.** Unplug your laptop completely or power your entire breadboard setup exclusively from an isolated **5V USB Power Bank**. This removes any risk of high-voltage mains shocks.
 - **Waveform Injection Testing:** Convert raw digital ECG waveforms from the MIT-BIH Database into .wav audio files. Plug an auxiliary cable from your laptop's headphone jack directly into the input pins of your breadboard's AD8232 sensor. Play the audio file to trick your prototype into believing it is attached to a real patient, allowing you to debug your hardware classification accuracy safely at your desk.
 - **The Event-Driven Test:** Verify that when a normal heartbeat audio file plays, the ESP32 draws nearly zero power (μA). When an arrhythmia waveform plays, the FPGA's GPIO pin must instantly transition from LOW to HIGH. This physical voltage jump must wake up the ESP32, prompt it to turn on its Bluetooth radio, and transmit the last 5 seconds of the anomaly data to your phone.
 - **The Hospital Parallel Trial:** In a controlled clinical setting with institutional approval, attach your battery-powered 3-lead electrode pads to a volunteer patient concurrently alongside the hospital's official, certified clinical ECG monitor. Use the hospital's monitor as your absolute ground truth to verify if your custom edge system successfully alerts you to real-world cardiac events in real time.
-

Project Intuition Checklists

Software Developer Mindset (Phase 1)

- *Am I designing for an infinite computer or a limited chip?* Every extra layer or filter I add in Python takes up physical block memory. I must keep the total network footprint under **15,000 bytes**.
- *Did I use QAT?* If I skip Quantization-Aware Training, my model will fail when compressed to INT8 on the hardware.

Hardware Engineer Mindset (Phase 2 & 3)

- *Where is the math happening?* The math is not running on a CPU core; it is happening across 8 dedicated physical hardware multiplier blocks on the ICE40 fabric.
- *Is my radio drawing power?* The ESP32 should remain cold and inactive 99% of the time, keeping overall current draw at minimal milliamp/microamp scales.

Clinical Tester Mindset (Phase 4)

- *Is the patient safe?* Double-check that the breadboard is completely disconnected from any wall outlets and running purely on battery power before placing electrodes on skin.